Commercial Airplane Multi-Engine Land Additional Rating Course

04/07/2025 (Rev 8)

Chapter 1 – Commercial Airplane Add-On Multi Engine Land Training Course Outline (TCO)

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Record of Revisions

REV NO.	DATE	INITIALS	LIST of CHANGES
ISSUE	11/23/2011	SP	
REV 1	03/13/2012	SP	
REV 2	08/03/2012	SP	
REV 3	04/02/2013	SP	
REV 4	11/04/2015	JL	
REV 5	05/17/2017	NS	
REV 6	05/17/2018	NS	
REV 7	05/25/20217	CG	Pg. 4- Changed format to record of revisions Pg. 1-30- Entire doc name change/ logo to L3harris " Pg. 5- removed reference to preamble Pg. 6- added DA42; added ref. to 141.39 for AC used. removed Mode C and Added ADSB to transponder, removed Autopilot req. Pg. 7,14 & 25- removed Lambert, Dugger and Gyuris added Crutchfield, respectively. Pg 8 – replaced Hurley with Hilmer as Chief Ground, added Thompson as Asst. Chief Ground Pg. 10 amended Course Overview "Training Syllabus". corrected Training Course to reflect 2 stages in flight. added NOTE for JU students Pg. 13- removed Computer Based Training (CBT) from Ground Training; added "Training Manager" as designee for lesson sequence. Pg. 14- removed Matt Lambert as Asst. Chief. Pg. 18-22 and 28- Updated Facility Diagrams. Pg. 24 added Frasca RTD Pg. 25- Updated Jacksonville Base Address (same facility address changed) Pg. 27- removed Aerosim AATD added Frasca RTD
Rev 8	04/07/2025	DK	Change all L3Harris references to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

FAA APPROVED

PAULINE AYN O'CONNOR

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Date: 2025.04.28 07:53:55 -04'00'

ORLANDO FSDO-SO15

List of Effective Pages

CAMEL ADD-ON Chapter 1					
PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	8	04/07/2025	2	8	04/07/2025
3	8	04/07/2025	4	8	04/07/2025
5	8	04/07/2025	6	8	04/07/2025
7	8	04/07/2025	8	8	04/07/2025
9	8	04/07/2025	10	8	04/07/2025
11	8	04/07/2025	12	8	04/07/2025
13	8	04/07/2025	14	8	04/07/2025
15	8	04/07/2025	16	8	04/07/2025
17	8	04/07/2025	18	8	04/07/2025
19	8	04/07/2025	20	8	04/07/2025
21	8	04/07/2025	22	8	04/07/2025
23	8	04/07/2025	24	8	04/07/2025
25	8	04/07/2025	26	8	04/07/2025
27	8	04/07/2025	28	8	04/07/2025
29	8	04/07/2025	30	8	04/07/2025

Revision System

Commercial Multi Engine Add-On Airplane Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial Multi Engine Add-On Airplane Syllabus* to ensure that all current procedures are being followed.

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ORLANDO FSDO-SO15

Aircraft

The aircraft that may be used in this course are

Piper PA-44 (4 place)

Multi-Engine Propeller Aircraft Land as applicable below All aircraft used in this course must:

All aircraft used in this course comply with FAR 141.39 and must:

1. Be registered as a civil aircraft of the United States and certified in the standard airworthiness category.
2. Be maintained and inspected in accordance with the requirements of 14 CFR Part 91.
3. Have engine power controls and flight controls easily accessible and that operate in a normal manner from each pilot station.

Required Aircraft Equipment

1. Transponder with Mode S
2. Equipped with ADS-B IN and OUT or OUT
3. MFD and PFD or standard instrument package
4. GPS or ADF

Personnel

Chief Flight Instructor

The Chief Flight Instructor designated for this course is designated in Webopss A006 and A007. The Chief Flight Instructor meets the requirements outlined in 14 CFR 141.35.

Assistant Chief Flight Instructors

The Assistant Chief Flight Instructor(s) designated for this course is found in the current Webopss A006 and A007.

The Assistant Chief Flight Instructor(s) designated for this course must meet the requirements outlined in 14 CFR 141.36 and will be designated by letter.

Flight Instructors

Each flight instructor assigned to this course must have completed Flight Instructor Training and be the holder of at least a Commercial Pilot Certificate with an Airplane category and Single Engine Land or Multi Engine Land class rating, as appropriate. In addition, each instructor must hold a Flight Instructor Certificate with an Airplane category and Single Engine class rating or Multi Engine class rating, as appropriate.

Chief Ground Instructor

The Chief Ground Instructor for this course is found in the current Webopss A006 and A007. The Chief Ground Instructor meets the requirements outlined in 14 CFR 141.35(e).

Whenever a Chief or Assistant Chief Ground Instructor is either undesignated or unavailable, the appropriate Chief or Assistant Chief Flight Instructor may assume those duties.

Assistant Chief Ground Instructors

The Assistant Chief Ground Instructor(s) designated for this course is found in the current Webopss A006 and A007.

The designated Assistant Chief Ground Instructor(s) for this course meets the requirements outlined in 14 CFR 141.36(e) and will be designated by letter.

Student Information

Course Enrollment

A person must hold the level of pilot certificate for the additional aircraft category and class rating for which the course applies prior to enrolling in the flight portion of an additional aircraft category or additional aircraft class rating course.

Requirements for Graduation

Crewmembers must satisfactorily complete the training outlined in this syllabus and be at least 18 years of age.

Lesson Description and Stages of Training

Each lesson is fully described within the syllabus, including the objectives, standards, and measurable units of accomplishment and learning. The stage objectives and standards are described at the beginning of each stage within the syllabus.

Tests and Checks

The syllabus incorporates stage checks and End of Course Tests in accordance with 14 CFR Part 141. These checks are given by the Chief Instructor, a designated Assistant Chief, or another designated Check Instructor as appropriate.

Course Overview

Course Objective

The student will obtain the knowledge, skill, and aeronautical experience necessary to meet the requirements for a Commercial Pilot AMEL Airplane.

Course Completion Standards

The student will demonstrate through written, oral, and flight tests that the knowledge, skill, and experience requirements necessary to obtain a Commercial Pilot Add-On AMEL are met.

Training Syllabus

The syllabus used will be the most current revision of the Acron Aviation Academy FAA Approved syllabus. (Chapters 2 and 3 of this TCO)

The training syllabus meets all curriculum requirements as prescribed by 14 CFR Part 141, Appendix I. It is composed of separate ground and flight courses to allow greater flexibility and to accommodate individual needs. The training courses may be taught either separately or concurrently.

Training Course

The Ground Training Course contains one (1) stage. The Flight Training Course contains two (2) stages.

NOTE

Students enrolled at Jacksonville University and completing a Bachelor of Science Degree in Aviation Management and Flight Operations receive full credit for all Ground School requirements for the following courses: Private ASEL, Instrument Airplane, Commercial Multi-Engine, CFI-A, and CFI-I.

Course Introduction

The Commercial AMEL Add-On Course utilizes the building-block theory of learning, which recognizes that each item taught must be presented based on previously learned knowledge and skills.

Course Elements

The Commercial AMEL Add-On Course utilizes separate flight and ground segments. The course includes the latest FAA instructor certification requirements and a maximum of student-oriented instruction. The syllabus and support materials not only provide necessary information, but also guide the student through the course in a logical manner.

Airplane Practice

Airplane practice must be conducted so that the student obtains the maximum benefit from each flight. Prior to each solo flight, an instructor will carefully instruct the student in the maneuvers to be performed during the flight and what is to be accomplished. This guidance will ensure that the student receives maximum benefit from the solo flight.

Use of Flight Training Devices

The Commercial AMEL Add-On allows, as an option, the use of approved flight training devices for those lessons designated in the Course Time Allocation Table under the Aircraft Type heading. The best results are obtained when a flight training device introduction is accomplished just prior to the flight lesson.

Preflight Briefing

The preflight briefing is an integral part of the learning process. During the preflight briefing, the PUI (Pilot under training) firstly brief the IP (Instructor Pilot). During this briefing the PUI will brief the IP on the possible hazards to the route of flight, review current weather, current personal minimums, and any other items of importance. Once the PUI has conducted their briefing the IP must review the items and challenge the information with specific questions. Once this is complete the IP will thoroughly brief the PUI on the scenario, and all the listed tasks and completion requirements. The "*Preflight Briefing*" should take a minimum of 15 minutes per flight/flight training device event.

Postflight Evaluation

The postflight evaluation is equally as important as the preflight briefing. Each scenario must be thoroughly debriefed, during which time the PUI will evaluate their performance and debrief the IP. The importance of this "*Self-Evaluation*" is to help the PUI learn to be critically honest of their own performance and to learn to come up with possible ways to correct any deficiencies without input from the IP. The IP will then debrief the PUI while emphasizing areas where the IP saw deficiencies, or where there was a notable difference in the "*Self-Evaluation*" by the PUI and the evaluation of the IP. Noticeable advancement should be apparent, and recommendations should be made for improvement, where appropriate. This action is a valuable instructional technique because it increases retention and helps to prepare the student for the next lesson. The "*Postflight Evaluation*" should take a minimum of 15 minutes per flight/flight training device event.

Student Stage Checks

Stage checks measure the student's accomplishments during each stage of training. The conduct of each stage check is the responsibility of the Chief Instructor. The authority to conduct these tests may be delegated to an appropriately designated and approved Assistant Chief Instructor or Check Instructor. This procedure provides close supervision of training and another opinion on the student's progress. The stage check also gives the Chief Instructor an opportunity to check the effectiveness of the instructors and their teaching methods.

An examination of the building-block theory of learning will show that it is extremely important for progress and proficiency to be satisfactory before the student enters a new stage of training.
Therefore, the next stage should not begin until the student successfully completes the current stage. Failure to follow this progression may defeat the purpose of the stage check and lead to overall course breakdown.

Ground Training

In accordance with 14 CFR Part 141, ground training is an integral part of instructor certification courses. The ground training syllabus in this course has been designed to meet this requirement.

The Ground Training Course may be presented to the student either as a formal classroom program or as an individual program. This allows for greater flexibility and accommodation to individual student needs.

Using the Ground Lessons

The ground lessons are generally divided into three sections--Lesson Introduction, Lecture, and Class Discussion. During the introduction, the instructor should outline the subject material to be covered during the training session, the objective for learning that information, and the performance standards necessary for successful lesson completion.

Textbooks

Prior to each ground lesson, the student should read and study the appropriate textbook section or chapter, as assigned. The appropriate textbook section or chapter will contain the information necessary to help the student complete the academic stages of the Commercial AMEL Add-On Course.

DVD Presentations

Following the ground lesson presentations, the student, as an option, may view DVD presentations. After the DVD presentation, the student should feel free to ask questions and obtain clarification of subject material viewed.

Final Examination

When all appropriate ground lesson presentations are complete, the student must take the Training Course Final Exam. Following the test, the instructor should assign the student review of appropriate subject areas. After a thorough review, the actual FAA Knowledge Test, if applicable, should be taken.

Commercial AMEL Add-On Training Course Implementation

The lesson sequence and content have been designed to provide the student with maximum training prior to the introduction of new maneuvers or procedures.

If any deviation from the normal lesson progression is desired, only the Chief Instructor, designated Assistant Chief or Training Manager for the course may grant approval.

NOTE

If the student is completing the ground training and flight training courses concurrently, the instructor must ensure that the student's flight training does not progress faster than the ground training.

Sanford Facility (SFB)

Acron Aviation Academy's Sanford base is located at the Orlando Sanford International Airport, 2700 Flightline Avenue, Sanford, FL 32773.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor for Acron Aviation Academy is found in Webopss A006 and A007.

Airport

The Orlando Sanford International Airport (SFB) is the main operations base for training. The airport has four (4) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

A federally operated control tower operates daily between 0630-2300 hours. The airport has ILS and GPS approaches.

Fuel and maintenance services are available daily between 0600-2300 hours.

Training Airports

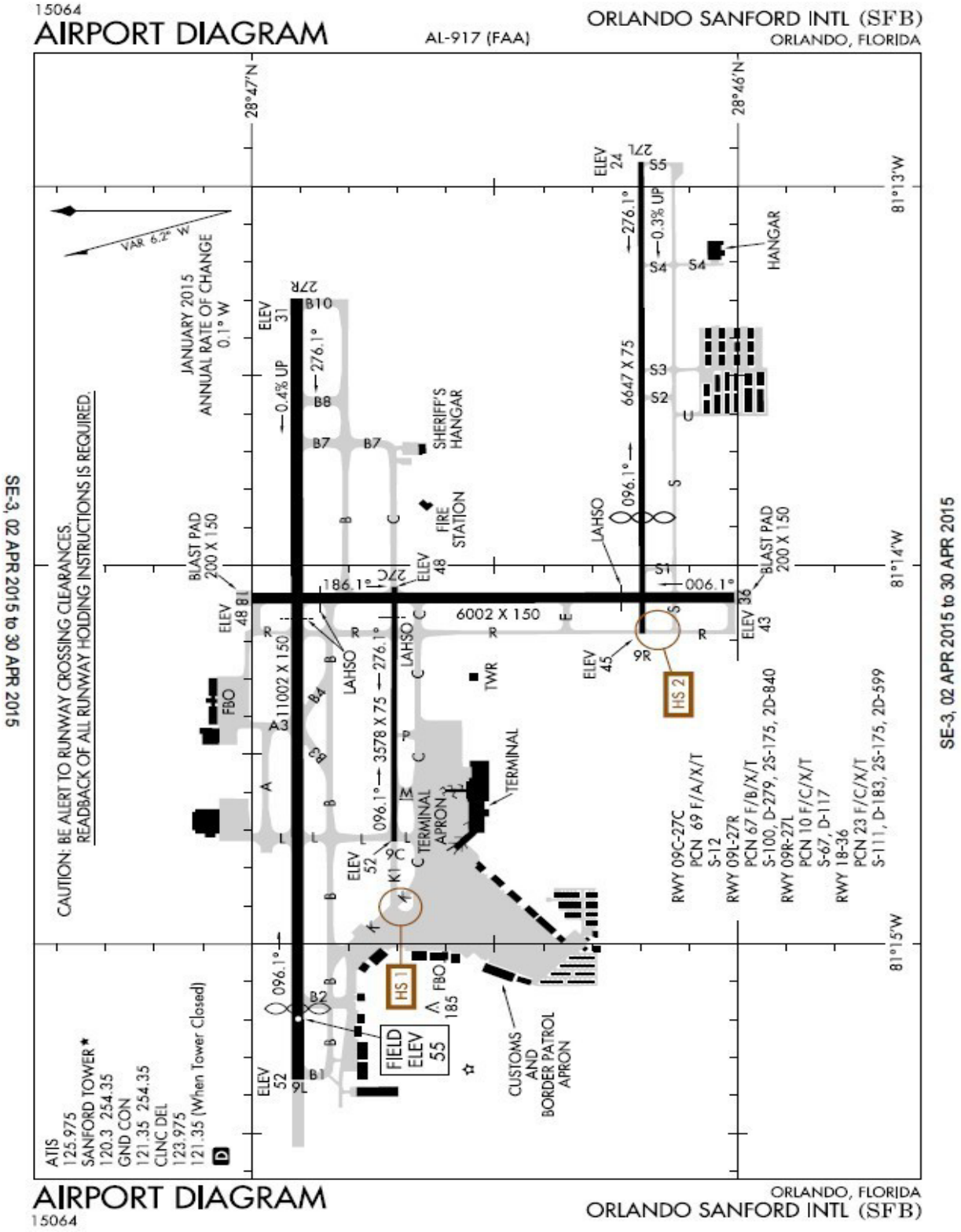
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy's flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Chart Supplement.

Sanford Airport Diagram

NOTE Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for preflight briefings, dispatch, and postflight briefings of all flight lesson activities.

The Operations Center includes a Planning Area and Briefing Rooms. The Flight/Ground Briefing Center provides additional space for briefings and small group instruction. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

Planning Area
This area is equipped with tables and chairs adequate for both students and instructors.

Telephones are available to provide communication to the nearest FAA Flight Service Station. In addition, weather stations are also available for preflight planning purposes.

Sanford Simulator Training Building

Acron Aviation Academy's Simulator Training Building is the focal point for all simulator and flight training device activity. Adequate areas have been provided for AATD and FTD training lessons.

The Simulator Training Building includes Briefing Rooms. Each briefing area is equipped with an erasable marker board as well as tables and chairs to accommodate students and instructors.

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Academics

The Academics facilities are in the Academics Building. This building houses classrooms, and the Learning Resource System.

Classrooms

Classrooms are equipped with tables and chairs to accommodate students. Each classroom contains at least one (1) chalkboard or erasable marker board of adequate size and one (1) projection screen.

Learning Resource Center

The Learning Resource Center (LRC) is equipped with tables and chairs to accommodate students. The LRC is available to students wishing to accomplish additional studying in a group or with a tutor. In addition, the LRC provides computer stations to students wishing to accomplish study using CBT materials, or by the viewing of DVD's.

Computer Testing Building

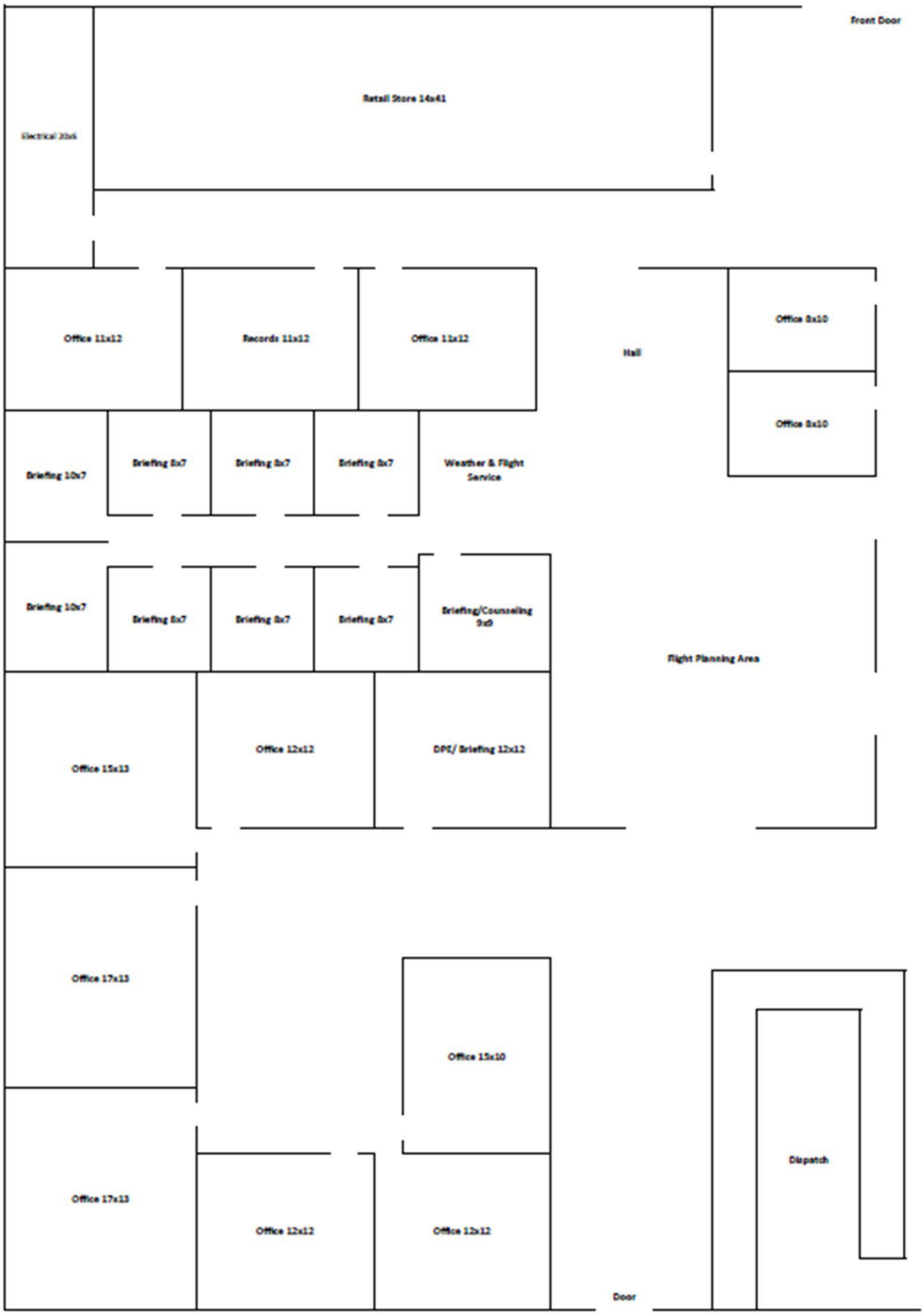
The Computer Testing Building is located at Bldg 41 2749 Flightline Ave, Sanford, Fl 32773. The Computer Testing Building has been set aside for the purpose of administering computerized FAA Knowledge Test Examinations.

NOTE

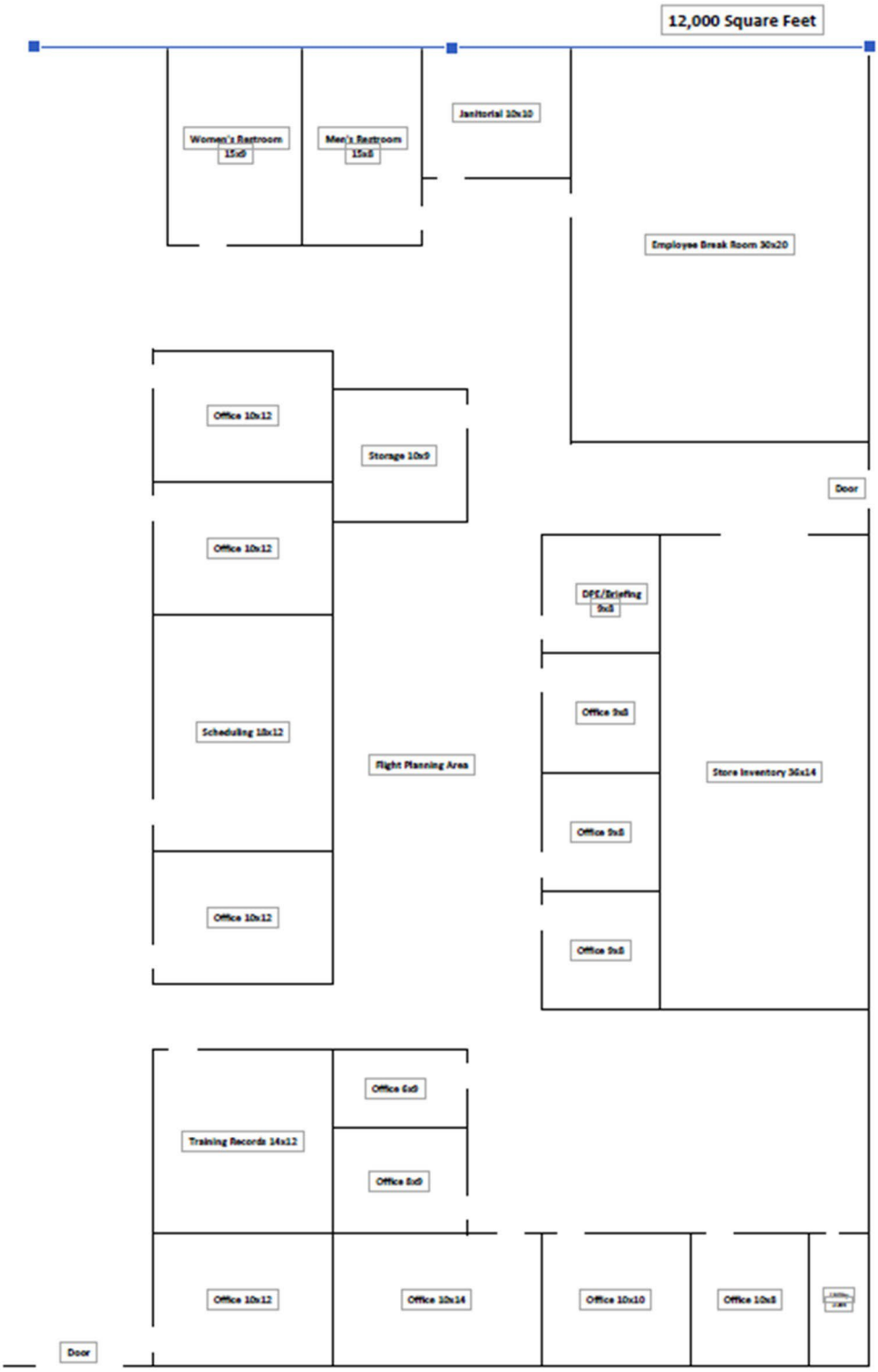
All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Building Diagrams

Sanford Operations Building – East Section

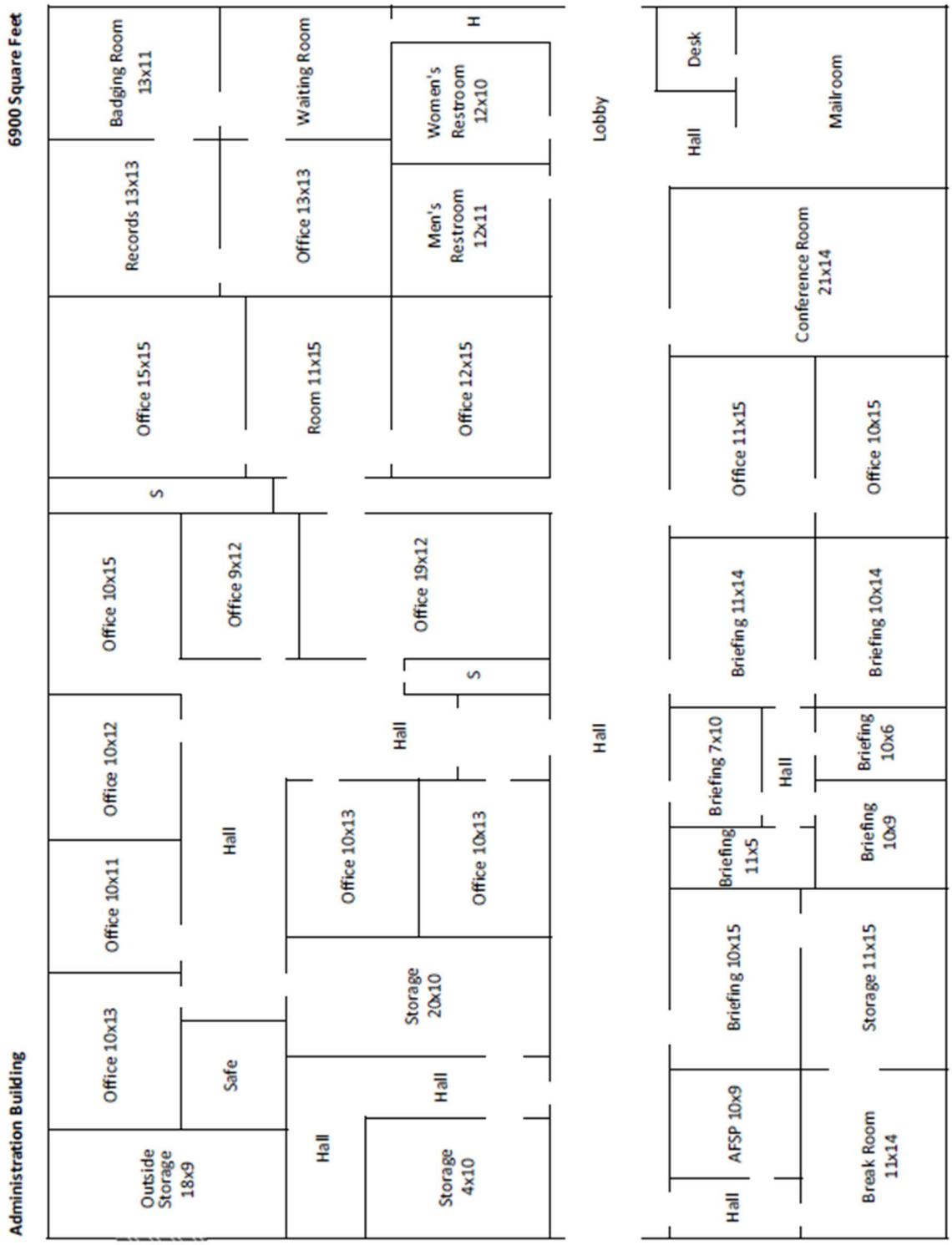


Sanford Operations Building – West Section

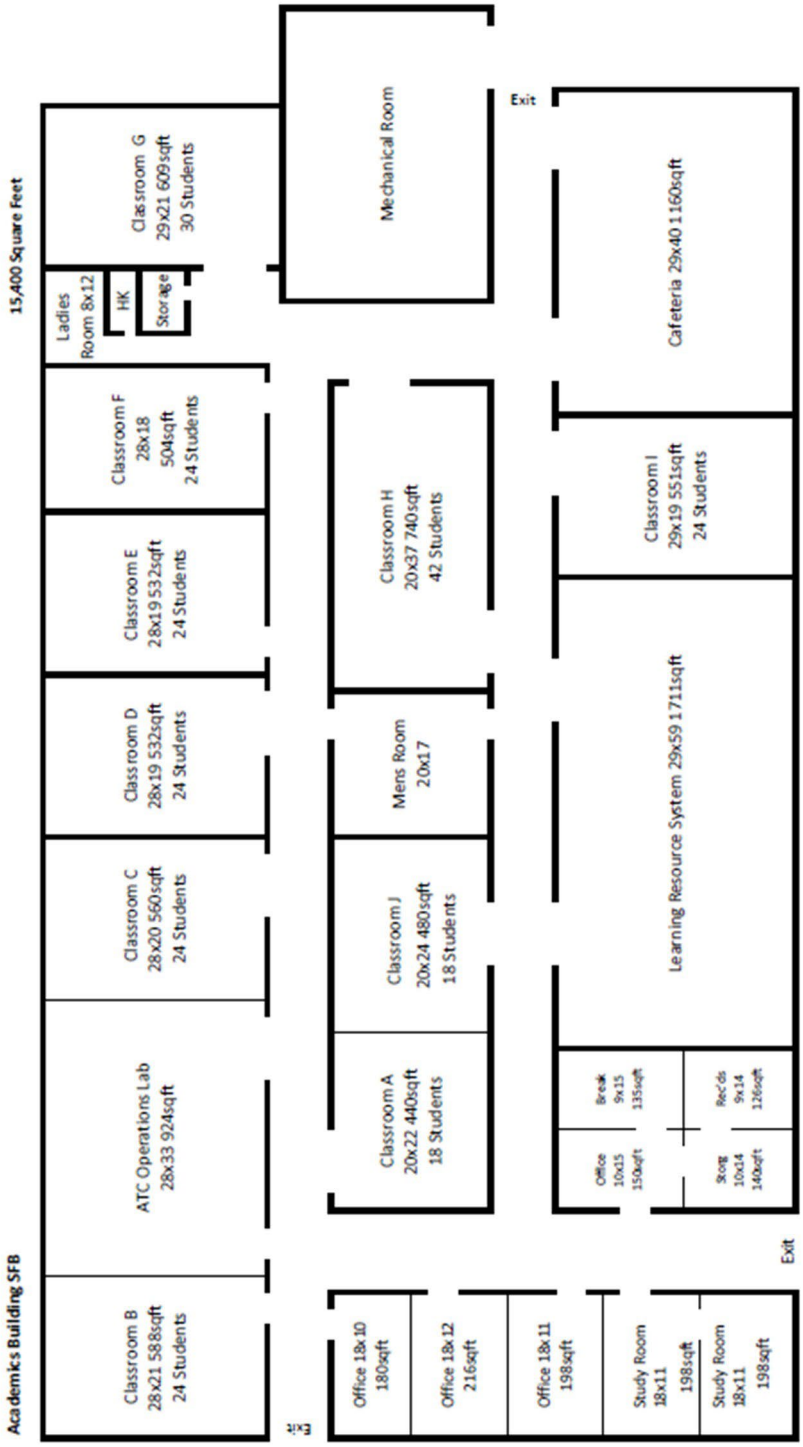


Sanford Administration Building

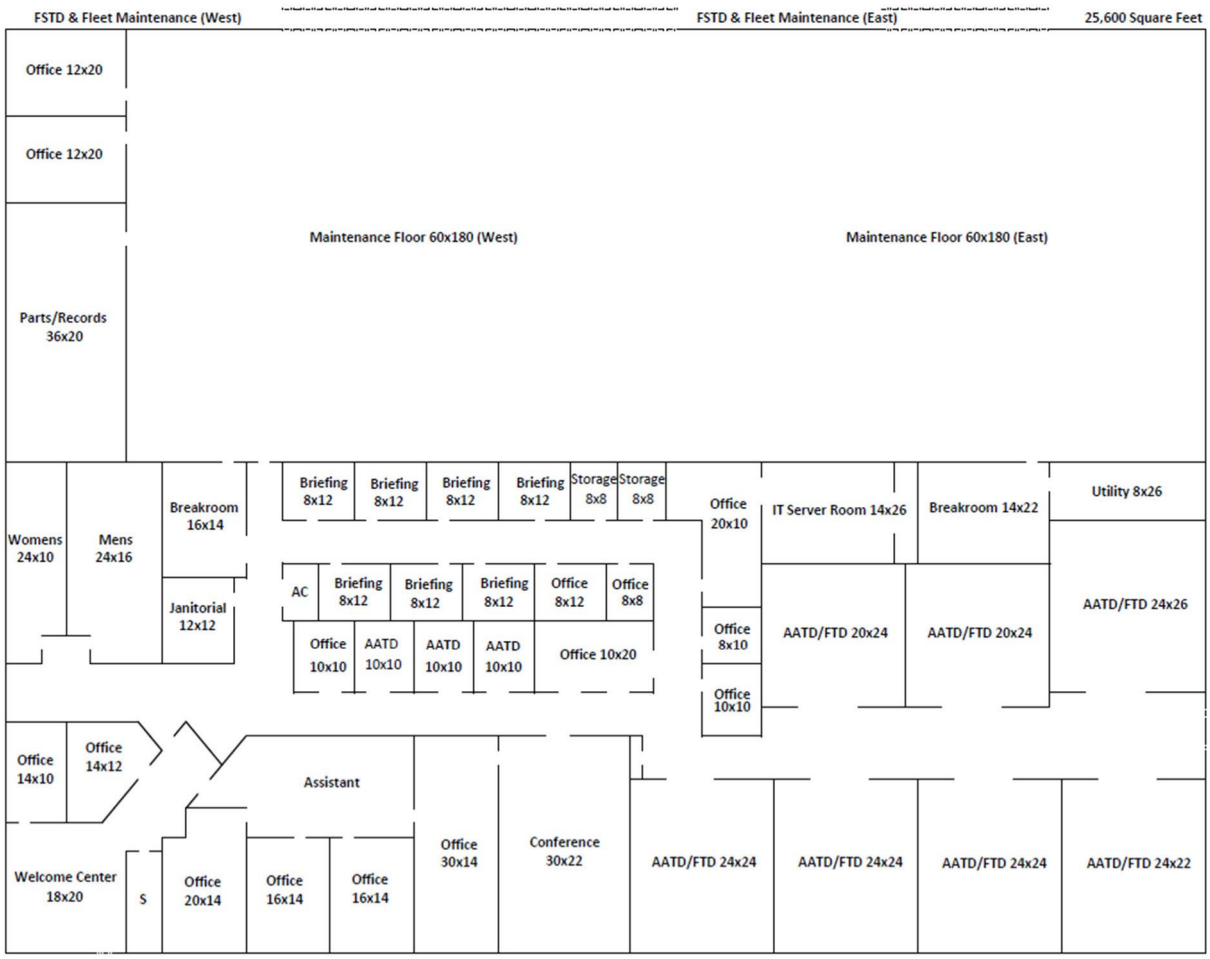
Note Building not currently used for academy operations



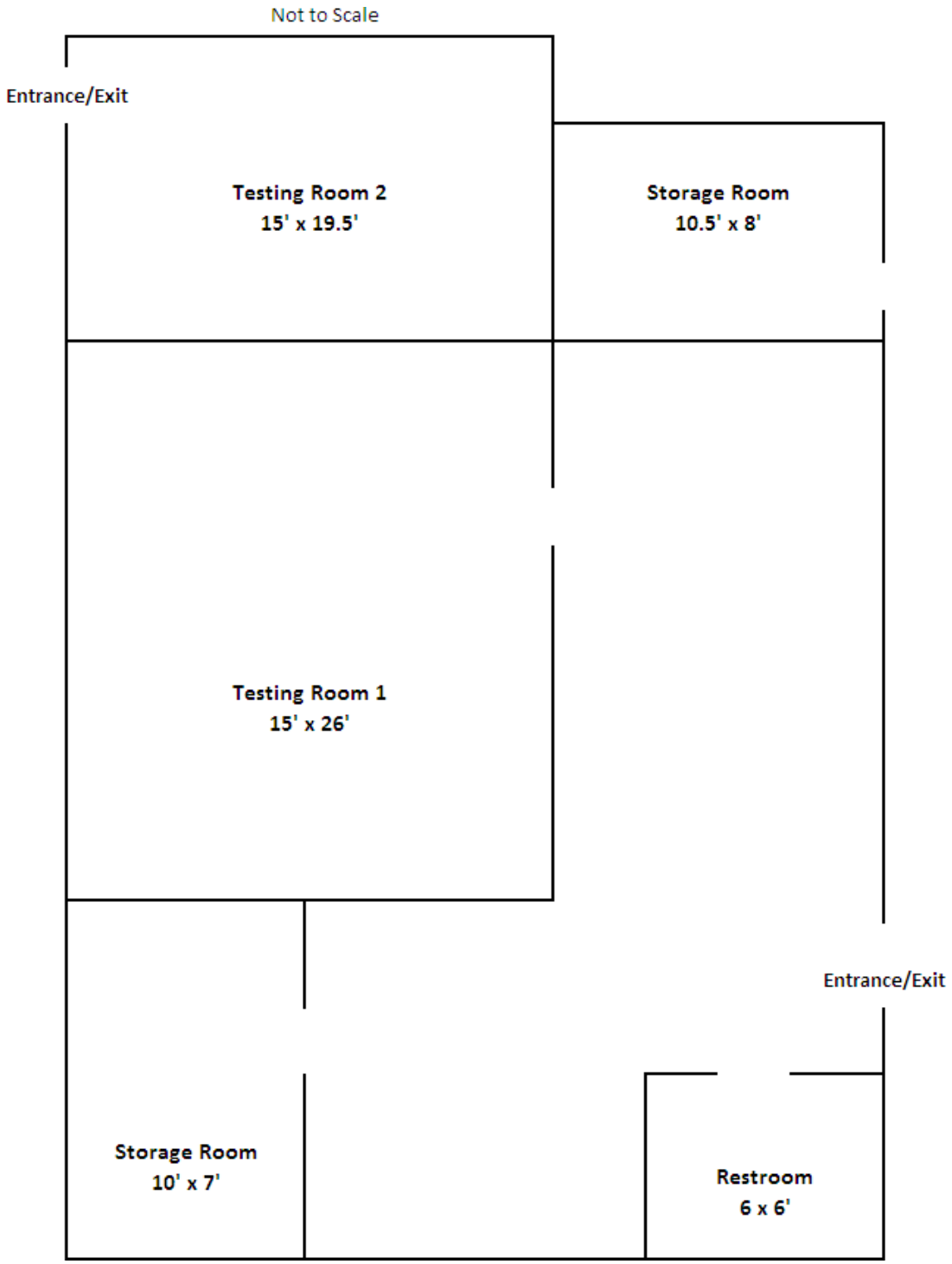
Sanford Academics Building



Sanford Simulator Training Building



Sanford Computer Testing Building



Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers with projectors, overhead projectors, Interactive Smart Boards, Diagrams and Charts
B. Mockups
   Large, presentation-type, E6B flight computer and Cockpit Procedures Trainer
C. Aircraft Components
   Flight instruments and aircraft engines on stands

In addition, for enhancement and reinforcement of training a variety of video tapes and computer- based training is available.

Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

1. Aerosim-Mechtronix Level 6 FTD
   The Aerosim-Mechtronix Level 6 flight training device (FTD) is an enclosed stationary trainer.
   equipped with an exact replica of the aircraft that meets FAA Level 6 certification criteria. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
2. Aerosim-Mechtronix Advanced Aviation Training Device (AATD)
   The Aerosim Model Aviator 100 Single Engine Land is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
3. The Frasca International Inc. TruFlite Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
4. The Redbird MCX Advanced Aviation Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically advanced, single, and multi-engine Airplane (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.
5. The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Jacksonville Facility (CRG)

Acron Aviation Academy's Jacksonville satellite base is located at the Jacksonville Executive At Craig Airport, 793 Amelia Earhart Drive, Jacksonville, FL 32225.

Chief and Designated Assistant Chief Flight Instructor

The Chief Flight Instructor and Assistant Chief Flight Instructor for Acron Aviation Academy is found in the current Webopss A006 and A007.

Airport

The Jacksonville Executive At Craig Airport (CRG) is the satellite base's main location for training. The airport has two (2) hard surface runways and meets the requirements of 14 CFR Part 141.38 for day and night flight operations.

An FAA contract control tower operates between 0600-2300 Monday through Friday and 0700- 2200 Saturday and Sunday.

The airport has ILS, VOR, and GPS approaches.

Fuel service is available from 0600-2200 hours daily. Maintenance service is available full time from 0700-1530 and on call 1530-2200 daily.

Training Airports

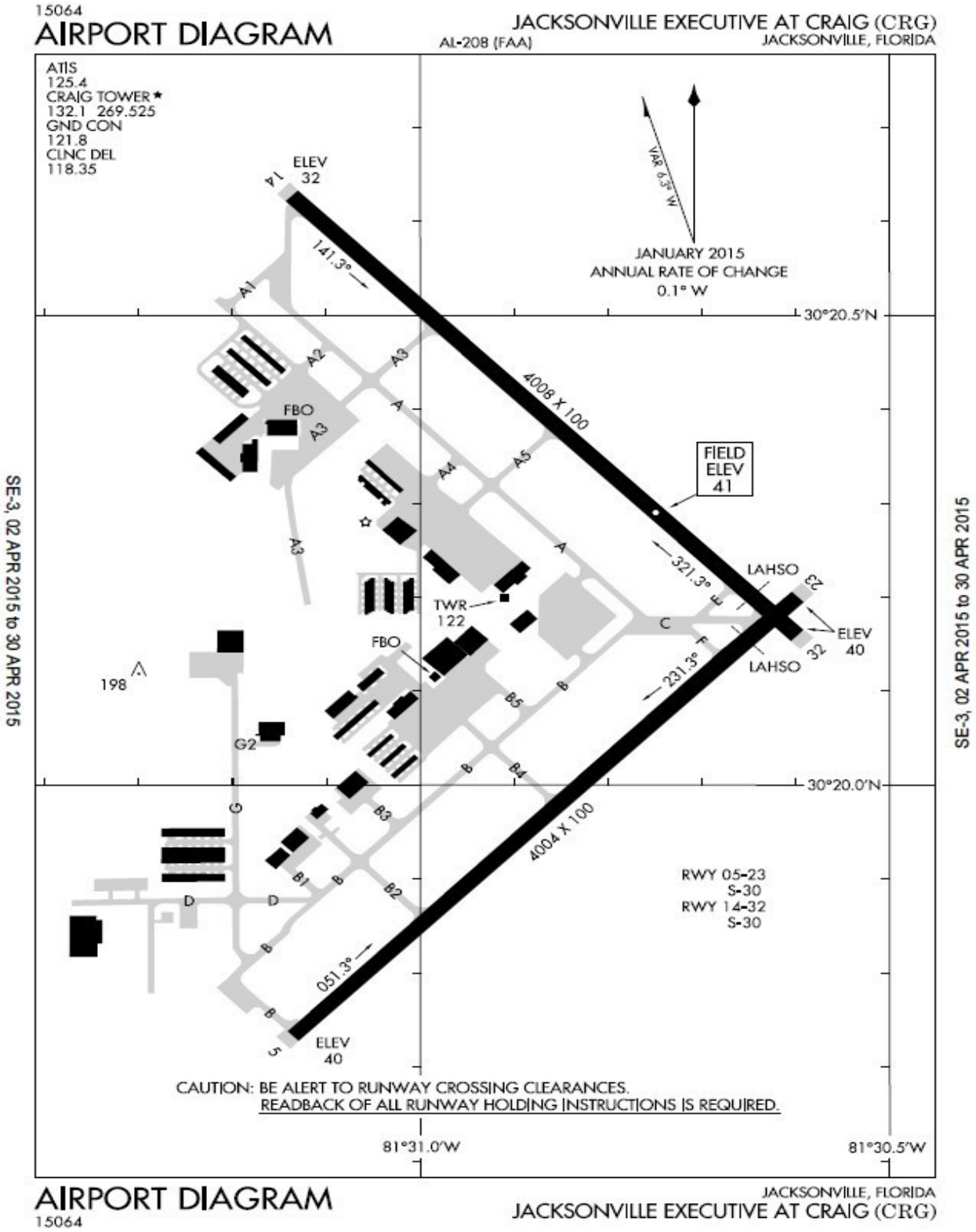
All training airports meet the requirements outlined in 14 CFR Part 141.38.

Direction is given to the Academy’s flight instructors through an Approved Airports list in supplemental documentation.

The Chief Flight Instructor or designated Assistant Chief Flight Instructor may approve any appropriate public use airport listed in the Airport/Facilities Directory.

Jacksonville/Crain Airport Diagram

Note Diagram not current and not to be used for navigation.



Operations Center

Acron Aviation Academy's Operations Center is the focal point for all flight lesson activity. Adequate areas have been provided for the dispatch and recovery of all flight lesson activity.

The Operations Center has adequate Pre-Post Flight Briefing Areas.

Pre-Post Flight Briefing Areas

Briefing Rooms 1 and 2 are equipped with tables and chairs adequate for both students and instructors. Briefing Room 1 may accommodate up to 16 students. Briefing Room 2 may accommodate up to 8 students. Briefing Room 2 may be reserved for FAA practical test oral examinations and stage checks as needed. All briefing areas are also equipped with erasable marker boards.

A telephone is available to provide communication to the nearest FAA Flight Service Station.

Simulator Hall

The Jacksonville facility has a Simulator Hall. The Sim Hall includes AATD simulators.

Equipment

Ground Training Aids

The training aids available for use, as appropriate, are as follows:

A. Audiovisual Aids
   Computers, Diagrams and Charts
B. Mockups
   Cockpit Procedures Trainer
C. Aircraft Components Flight instruments

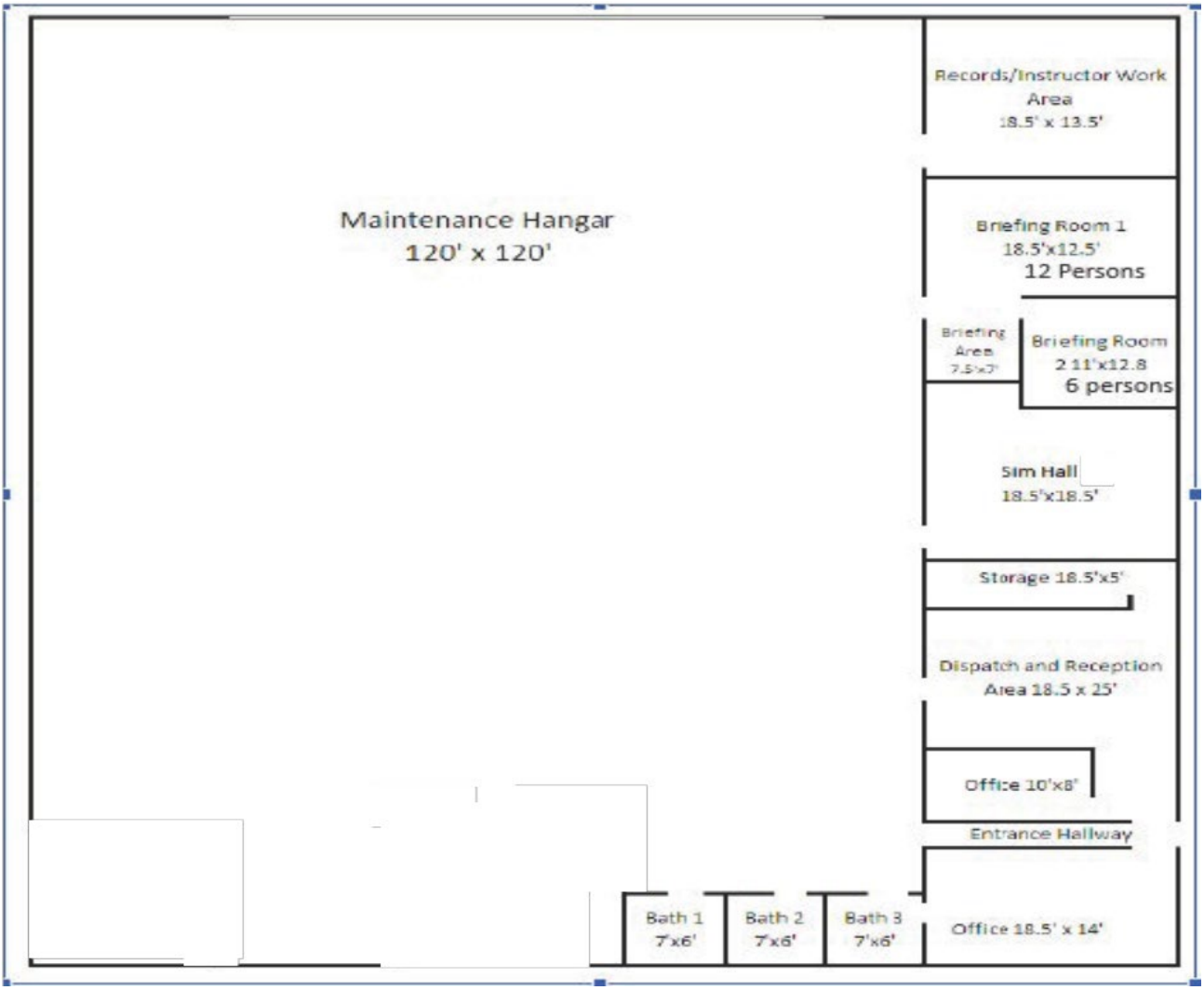
In addition, for enhancement and reinforcement of training a variety of video tapes and computer- based training is available.

Flight Training Devices

The Academy's flight training devices/approved aviation training devices meet the requirements specified in 14 CFR Part 141.41 (b)/14 CFR Part 141.41 (c) respectively, and may be used to meet the total simulated flight time allowed in 14 CFR Part 141, Appendices B through G and Appendix I.

-The Frasca International Inc. RTD Training Device (AATD) is a stationary trainer equipped with the necessary flight controls to simulate flight around the three axis of an airplane. It is equipped with the engine instruments and controls necessary for operating a Technically Advanced, single engine Airplane or multi engine (TAA). All the required equipment for VFR and IFR flight as per 14 CFR 91.205 (b) and (d) is provided or simulated.

Operations Center Diagram



Academics

The Academics facilities are located on the campus of Jacksonville University. Located at 2800 University Blvd. N., Jacksonville, FL 32211.

Classrooms

Classroom D-153 in the Davis College of Business Building may accommodate 41 students. Depending on the size of the class and the scheduling needs of the college other rooms may be assigned.

Ground Training Aids

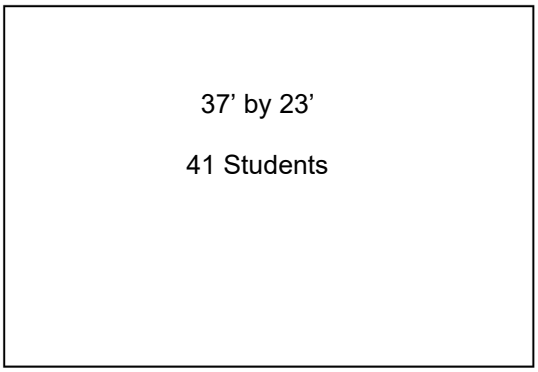
PC/VCR/DVD/Internet with overhead projection capability
Overhead projector
Computer-based training
Flight instruments
Static displays of aircraft engines and parts Aircraft models
Giant E6B White board

NOTE

All rooms and areas conform to local building, sanitation, and health codes. All rooms are designed and located so that students will not be distracted by the instruction conducted in other rooms or by other flight and maintenance operations on the airport.

Jacksonville University Classroom Diagram

Davis College of Business Room D-153



Chapter 2-Commercial Multi-Engine (AMEL) Add-On Ground Syllabus

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Revision System

Commercial AMEL AR course revisions will be issued in consecutive numbers and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial AMEL AR course* to ensure that all current procedures are being followed.

REV NO.	DATE	INITIALS	LIST of CHANGES
Issue	07/16/07	RM	
REV 1	05/15/08	RM	
REV 2	01/12/09	RH	
REV 3	12/10/09	RH	
REV 4	03/13/129	RH	
REV 5	05/17/17	NS	
REV 6	05/17/18	RH	
REV 7	01/25/2021	CG	**Pg. 2-** changed format to Record of Revisions **Pg. 1-23-** Entire doc.- Name Change/ logo to "L3Harris" **Pg. 5-** edited TAT, lesson 16, to show "Review/ Final Exam" and corrected Totals to 27.0 and 3.0
REV 8	04/07/2025	DK	Changed all L3Harris references to Acron

After revision insertion, enter the date the revision is incorporated and initial in the appropriate column.

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List of Effective Pages

PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	8	04/07/2025	2	8	04/07/2025
3	8	04/07/2025	4	8	04/07/2025
5	8	04/07/2025	6	8	04/07/2025
7	8	04/07/2025	8	8	04/07/2025
9	8	04/07/2025	10	8	04/07/2025
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13	8	04/07/2025	14	8	04/07/2025
15	8	04/07/2025	16	8	04/07/2025
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19	8	04/07/2025	20	8	04/07/2025
21	8	04/07/2025	22	8	04/07/2025
23	8	04/07/2025	23	8	04/07/2025

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CAMEL AR Course Objective

The objective of this course is to introduce the Pilot Under Instruction (**PUI**) to the Commercial AMEL Additional Rating aviation-training environment. The **PUI** will study all aeronautical knowledge areas required for a Commercial AMEL Pilot Certificate.

The instructor will conduct training using the following equipment:

Classroom
Computer
Projector Whiteboard

Reference Textbooks for this course include (Not all inclusive) :

14 CFR 1	AFM	POH/AFM
AC 60-28	FAA-H-8083-3	NOTAMS
AC 61-136	FAA-H-8083-15	
AC 91-74	FAA-H-8083-16	
AC 91.21-1	FAA-H-8083-25	
AC 120-108	FAA-ACS	

CAMEL AR Course Completion Standards

At the completion of this course, the **PUI** will demonstrate to the instructor a basic understanding of all aeronautical knowledge areas required for a Commercial AMEL Pilot Certificate. The **PUI** will be able to accomplish the objectives of the course to the appropriate level of performance, as outlined in the examination completion standards.

Suggested Course Time Allocation Table

Lesson		Training	Exam
1	Federal Aviation Regulations/NTSB	2.0	
2	Certificates and Documentation	1.0	
3	Advanced Aerodynamics	3.0	
4	High Altitude Operations	1.0	
5	Night Operations	1.0	
6	Weight and Balance	1.0	
7	Multi Engine Performance	2.0	
8	Multi Engine Systems	1.0	
9	Multi Engine Operations	3.0	
10	Multi Engine Maneuvers	1.0	
11	Engine Out Aerodynamics	2.0	
12	Engine Failure	2.0	
13	Engine Out Maneuvers	2.0	
14	Multi Engine ADM	1.0	
15	Airworthiness Requirements	1.0	
16	Review and Final Exam	0.0	3.0
CAMEL AR Course Totals		**24.0**	**3.0**

Course Minimum 15

Note: The individual times shown on this table are for instructor/student guidance only. They are not mandatory for each individual ground school lesson.

Ground School Lesson 1: FEDERAL AVIATION REGULATIONS/NTSB

Lesson Objective

During this lesson, the **PUI** will have an introduction to Commercial Piloting regulations/NTSB.

Lesson Content

Part 61 Part 91 Part 119 Part 121 Part 135 Part 141	
General Initial Notification of Aircraft Accidents, Incidents, and Overdue Aircraft Preservation of Aircraft Wreckage, Mail, Cargo, and Records Reporting of Aircraft Accidents, Incidents, and Overdue Aircraft	

Completion Standards

The lesson shall be complete when, by oral review, the PUI displays an understanding of the material covered.

Ground School Lesson 2: CERTIFICATES AND DOCUMENTS

Lesson Objective

During this lesson, the **PUI** will study certificates and documents.

Lesson Content

Commercial Pilot Certificate Medical Certificate Pilot Logbook Airworthiness and Registration POH/AFM Weight and Balance Data Equipment List	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 3: Basic AERODYNAMICS

Lesson Objective

During this lesson, the **PUI** will study airplane aerodynamics and the principles of flight.

Lesson Content

Lift Drag Thrust Weight and Load Factor The V-G Diagram Aircraft Stability Aerodynamics and Flight Maneuvers Stall and Spin Awareness	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI **displays** an understanding of the material covered.

Ground School Lesson 4: HIGH ALTITUDE OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study high altitude operations.

Lesson Content

Oxygen Requirements Physiological Hazards Pressurization and Oxygen Systems Aviators Oxygen Care of Oxygen Bottles Rapid Decompression Cabin Pressurization Concepts Cabin Pressurization Systems	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 5: NIGHT OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study night operations.

Lesson Content

Night Vision Night Illusions Interior Lights Flashlight Night Preflight Engine Starting/Aircraft Lights Taxi Orientation Takeoff and Climb In-flight Orientation Night BAI Night Emergencies Traffic Patterns Approaches and Landings Go-Around	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 6: WEIGHT & BALANCE

Lesson Objective

During this lesson, the **PUI** will study the factors affecting weight and balance.

Lesson Content

Weight Terms Effects of Weight Maneuvering Speed Center of Gravity Normal Loading Effects of Forward or Aft CG Weight Shifting Transport Loading	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 7: MULTI ENGINE PERFORMANCE

Lesson Objective

During this lesson, the **PUI** will study the factors affecting airplane performance. The significance and effects of exceeding aircraft performance limitations will be discussed.

Lesson Content

Performance Terms Power Required Power Required Engine Out Performance Data Usage Atmospheric Conditions V-Speeds Using the Performance Charts	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 8: MULTI ENGINE AIRCRAFT SYSTEMS

Lesson Objective

During this lesson, the **PUI** will study aircraft systems and associated failures.

Lesson Content

Airframe Flight Control System Engines Propellers Engine Controls Fuel System Electrical System	
Landing Gear Brake System Vacuum System Pitot Static System Environmental System Instrument Panel Cabin Features Stall Warning ELT	

Completion Standards

The lesson shall be complete when, by oral examination, the PUI displays an understanding of the material covered.

Ground School Lesson 9: MULTI-ENGINE OPERATIONS

Lesson Objective

During this lesson, the **PUI** will study multi engine commercial pilot operations.

Lesson Content

Checklist Usage Preflight Procedures Taxiing Takeoffs Cruise Descent Landings Go-Arounds	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 10: MULTI-ENGINE MANEUVERS

Lesson Objective

During this lesson, the **PUI** will study multi engine commercial pilot maneuvers.

Lesson Content

Steep Turns Slow Flight Stalls Spin Awareness Emergency Descent	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 11: ENGINE-OUT AERODYNAMICS

Lesson Objective

During this lesson, the **PUI** will study airplane multi engine aerodynamics and emergency checklist.

Lesson Content

Yaw, Roll, and Pitch Critical Engine Theories Aircraft Control Windmilling vs. Feathered Propeller Zero Sideslip Performance Loss Vmc Factors Emergency Checklist	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 12: ENGINE FAILURES

Lesson Objective

During this lesson, the **PUI** will study engine failures in multi-engine aircraft.

Lesson Content

Recognizing an Engine Failure Identify, Verify, and Decide Troubleshoot Feather Securing the Inoperative Engine Monitoring Training Considerations	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 13: ENGINE-OUT MANEUVERS

Lesson Objective

During this lesson, the **PUI** will study engine failure maneuvers in multi-engine aircraft.

Lesson Content

Failure Before Takeoff Failure After Takeoff Failure Enroute Vmc Demonstration Vyse Demonstration Single Engine Landing and Go-Around	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 14: MULTI-ENGINE ADM

Lesson Objective

During this lesson, the **PUI** will study aeronautical decision making in the multi engine environment.

Lesson Content

Decision Making Process PIC Communication Resources Management Awareness CFIT Poor Judgment	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 15: AIRWORTHINESS REQUIREMENTS

Lesson Objective

During this lesson, the **PUI** will study airworthiness requirements.

Lesson Content

Day/Night VFR/IFR Equipment Requirements MEL Special Flight Permit Airworthiness Directives Compliance Records Maintenance and Inspections Record Keeping	

Completion Standards

The lesson shall be complete when, by oral examination, the **PUI** displays an understanding of the material covered.

Ground School Lesson 16: STAGE I REVIEW/EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on comprehension of the academic material presented in Stage I.

Lesson Content

Stage I Review Stage I Exam	

Completion Standards

The **PUI** should complete the Commercial Stage II Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

Ground School Lesson 17: FINAL EXAM

Lesson Objective

During this lesson, the **PUI** will be tested on their comprehension of the academic material presented in this course.

Lesson Content

Final Exam	

Completion Standards

The **PUI** should complete the Commercial End-of-Course Exam with a minimum passing score of 80% and the instructor should review each incorrect response to ensure complete understanding.

The student will demonstrate through written tests and school records that the aeronautical knowledge necessary to pass the FAA Commercial Pilot Airplane Knowledge Examinations has been acquired. The student must meet the requirements specified in 14 CFR Part 141, Appendix D.

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Chapter 3 – Commercial Multi-Engine Add-On Flight Syllabus

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Record of Revisions

REV NO.	DATE	INITIALS	List Of Changes
REV 2	03/13/2012	SP	
REV 3	08/03/2012	SP	
REV 4	11/04/2015	JL	
REV 5	05/17/2017	NS	
REV 6	01/25/2021	CG	• Name change to L3Harris • Edited all reference to L3 Commercial Training Solutions to “L3Harris” • Changed equipment for ground trainer to be “AATD/FTD” • Replaced reference from PTS to ACS • Reclassified DFT (Dual FTD) to TD (Training Device) • Edited NOTE pg.6 to move AATD/FTD events to airplane as needed • Pg.2- Changed record of revisions with new format
REV 7	04/07/2025	DK	Changed all L3Harris references to Acron

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List of Effective Pages

Commercial AMEL Syllabus					
PAGE	REVISION	DATE	PAGE	REVISION	DATE
1	7	04/07/2025	2	7	04/07/2025
3	7	04/07/2025	4	7	04/07/2025
5	7	04/07/2025	6	7	04/07/2025
7	7	04/07/2025	8	7	04/07/2025
9	7	04/07/2025	10	7	04/07/2025
11	7	04/07/2025	12	7	04/07/2025
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23	7	04/07/2025	24	7	04/07/2025
25	7	04/07/2025	26	7	04/07/2025
27	7	04/07/2025	28	7	04/07/2025
29	7	04/07/2025	30	7	04/07/2025
31	7	04/07/2025	32	7	04/07/2025
33	7	04/07/2025	34	7	04/07/2025

Revision System

Commercial AMEL Add On Syllabus revisions will be issued in consecutive numbers; and each page of the new revision will contain the revision number and the date the revision was issued.

It is the responsibility of all affected personnel to periodically review the *Commercial AMEL Add On Syllabus* to ensure that all current procedures are being followed.

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Suggested Course Time Allocation Table

STAGE NUMBER	LESSON NUMBER	SCHEDULED TIME	PRE BRIEF	POST BRIEF	AATD/FTD	AIRCRAFT	DUAL	NIGHT	INSTRUMENT	STAGE CHECK ORAL	STAGE CHECK FLIGHT	EQUIPMENT TYPE
1	1-TD	1.2	0.2	0.25	1.0		1.0					AATD/FTD
1	2-DL	2.0	0.2	0.25		1.5	1.5					Airplane Multi-Engine Land
1	3-DXC	2.5	0.2	0.25		2.0	2.0					Airplane Multi-Engine Land
1	4-TD	1.7	0.2	0.25	1.5		1.5					AATD/FTD
1	5-DXCN	2.5	0.2	0.25		2.0	2.0	2.0	1.5			Airplane Multi-Engine Land
1	6-STO	2.0	0.2	0.25						2.0		Ground
1	7-STF	2.5	0.2	0.25		2.0	2.0		1.5		2.0	Airplane Multi-Engine Land
2	8-TD	1.8	0.2	0.25	1.5		1.5		1.5			AATD/FTD
2	9-TD	1.8	0.2	0.25	1.5		1.5		1.5			AATD/FTD
2	10-DL	3.0	0.2	0.25		2.5	2.5					Airplane Multi-Engine Land
2	11-DL	3.0	0.2	0.25		2.5	2.5					Airplane Multi-Engine Land
2	12-STO EOC	2.5	0.2	0.25						2.5		Ground
2	13-STF EOC	3.0	0.2	0.25		2.5	2.5		1.0		2.5	Airplane Multi-Engine Land
		Totals	**3.2**	**3.25**	**5.50**	**15.00**	**20.50**	**2.00**	**7.00**	**4.50**	**4.50**	

Course Minimums		
Minimum Briefing Time	**Multi Engine Instrument**	**Multi Engine/Complex Training**
6.9	5	15

Note: The individual times shown on this table are for Instructor/Student guidance only. They are not mandatory for each flight.

Stage 1

Stage Objective

The objective of this stage is to introduce the Pilot Under Instruction (**PUI**) to aircraft operations in a multiengine aircraft and advanced flight maneuvers listed in the FAA Commercial Pilot (AMEL) Airman Certification Standards. The **PUI** will be introduced to cross-country missions and w i l learn the skills necessary to perform basic and advanced maneuvers during the enroute phase of flight while experiencing real-world Lessons. The **PUI** will continue using aeronautical d e c i s i o n - making skills that will set the foundation for all future aviation training.

The **PUI** will conduct training in the following equipment:

Training Device (TD)
Advanced Aviation Training Device **(AATD)**
Multi-Engine Flight Training Device **(MFTD)**

Aircraft
Airplane Multi-Engine Land **(AMEL)**

Stage Completion Standards

At the completion of this stage, the PUI **will** demonstrate to the Instructor Pilot (**IP**) an understanding of cross-country operations and multiengine commercial flight maneuvers. The **PUI** will be able to accomplish the objectives of Stage 1 to the appropriate level of performance, as outlined in the Lesson completion standards.

NOTE

*In the event an **AATD/ FTD** is not available, the training event prescribed can be conducted in the aircraft with the approval from the Chief Instructor/ Asst. Chief or designee.*

Lesson 1-TD

Equipment

AATD/FTD

Lesson Objective

This Lesson provides an introduction of the procedures and skills needed in a multiengine aircraft.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Cockpit Familiarization	
Flight Controls	Introduction
Engine Controls	Introduction
Checklist Usage	
Normal Procedures Checklist	Introduction
Flows	
Cockpit Flows	Introduction
Engine Start	
General Starting Procedures	Introduction
Taxi Operations	
Basic Aircraft Control	Introduction
Nose Wheel Steering	Introduction
Differential Braking	Introduction
Before Takeoff Check	
Engine Check	Introduction
Instrument Check	Introduction
Takeoff and Climb Procedures	
Normal Takeoff and Climb	Introduction
Climbs	
Constant Rate Climbs	Introduction
Constant Airspeed Climbs	Introduction
Operation of Systems In-Flight	
Powerplant Operation	Introduction
Use of Constant Speed Propeller	Introduction
Use of Retractable Landing Gear	Introduction
Use of Cowl Flaps	Introduction
Mixture Leaning including use of EGT	Introduction
Slow Flight	
Pitch/Airspeed Control	Introduction
Power/Altitude Control	Introduction
Climbs/Turns/Descents	Introduction

Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls	 Introduction Introduction Introduction
Steep Turns Steep Turns	 Introduction
Descents Constant Rate Descents Constant Airspeed Descents	 Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Go-Around/Rejected Landing	 Introduction Introduction
Postflight Operations Postflight Debrief	 Introduction
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees and altitude will be maintained within +/- 200 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 2-DL

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This lesson provides a review of the procedures and skills needed in a multiengine aircraft and an introduction to Short-Field landings.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Introduction
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review
Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review
Turns	
Standard Rate Turns	Review
Compass Turns	Review

Steep Turns Steep Turns	 Review
Slow Flight Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Review Review Review
Stalls Power-Off Stalls Power-On Stalls	 Review Review
Descents Constant Rate Descents Constant Airspeed	 Review Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-Field Landing Go-Around/Rejected Landing	 Review Introduction Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 20 degrees and altitude will be maintained within +/- 200 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 3-DXC

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This lesson provides a review of the procedures and skills needed in a multiengine aircraft and fulfills the requirements of Part 141, Appendix D 4(b)(2) iii and Part 61.129(b)(3) iii. The **PUI will** plan and execute a multi segment day VFR cross-country at least 100 NM from the airport of departure and a minimum of two hours in duration. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Preflight Procedures	
Lesson Planning	Review
Certificates and Documents	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Avionics	
GPS Setup and Usage	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review

Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review
Turns	
Standard Rate Turns	Review
Compass Turns	Review
Steep Turns	
Steep Turns	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
VFR Navigation	
E6B Usage	Review
Diversion Procedures	Review
Cross Country Flight	
Landing at an Airport at least 100 nm distance	
Cross Country at least 2 hr. duration	
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-Field Landing	Review
Go-Around/Rejected Landing	Review
Postflight Operations	
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	
Special Emphasis Areas (If Necessary)	Review

Completion Standards

At the completion of this lesson, the **PUI** will have completed a Day VFR cross country consisting of 100 nm straight line distance and 2.0 hours or more in duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table.
Heading will be maintained within +/- 15 degrees and altitude will be maintained within +/- 150 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 4-TD

Equipment

AATD/FTD

Lesson Objective

This lesson provides a review of the procedures and skills needed in a multiengine aircraft. During this lesson the **PUI** will be introduced to emergency operations in a multiengine aircraft. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the I P at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Principles of Flight--Engine Inoperative	Introduction
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Introduction
Emergency Procedures Checklist	Introduction
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Introduction
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Aborted Takeoff	Introduction
Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review

Emergency/Abnormal Procedures Engine Failure During Takeoff Before Vmc (Simulated) Engine Failure After Lift-Off (Simulated) Systems and Equipment Malfunctions Emergency Descent	 Introduction Introduction Introduction Introduction
Steep Turns Steep Turns	 Review
Multiengine Operations Maneuvering With One Engine Inoperative Vmc Demonstration Demonstrating the Effects of Various Airspeeds and Configurations During Engine Inoperative Performance	 Introduction Introduction Introduction Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Inoperative Engine Approach & Landing Procedures Go-Around/Rejected Landing	 Review Introduction Review
Postflight Operations Postflight Debrief	 Review
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 15 degrees and altitude will be maintained within +/- 150 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 5-DXCN

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This night lesson provides a review of the procedures and skills needed in a multiengine aircraft and fulfills the requirements of Part 141, Appendix D 4(b)(2) iv and Part 61.129(b)(3) iv. During this lesson the **IP** will continue the introduction of emergency operations in a multiengine aircraft. At the completion of this Lesson, the **PUI** will have completed a Night VFR cross country consisting of 100 nm straight line distance and 2.0 hours or more in duration.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Principles of Flight--Engine Inoperative	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review

Emergency/Abnormal Procedures	
Engine Failure During Takeoff Before Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Systems and Equipment Malfunctions	Review
Mixture Leaning including use of EGT	Review
Emergency Descent	Review
Multiengine Operations	
Maneuvering With One Engine Inoperative	Review
Vmc Demonstration	Review
Demonstrating the Effects of Various Airspeeds and Configurations	Review
During Engine Inoperative Performance	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Climbs/Turns/Descents	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Cross Country Flight	
Landing at an Airport at least 100 nm distance	
Cross Country Total Distance at least 100 nm	
Steep Turns	
Steep Turns	Review
Approach and Landing Procedures	
Short-field Landing	Review
Landing With an Inoperative Engine (Simulated) Go-	Review
Around/Rejected Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	
Special Emphasis Areas (If Necessary)	Review

Completion Standards

At the completion of this lesson, the **PUI** will have completed a Night VFR cross country consisting of 100 nm straight line distance and 2.0 hours or more in duration. The **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table.
Heading will be maintained within +/- 10 degrees and altitude will be maintained within +/- 100 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 6-STO–Stage Check (Oral)

Equipment

Briefing Area

Lesson Objective

The lesson will be conducted by the Chief/Assistant Chief or Designated Check Pilot.
The **PUI** will use SRM. At the end of the lesson, the **PUI** will meet the applicable PTS for a Commercial Pilot AMEL. This lesson provides a review of certificates and documents; airworthiness requirements; performance and limitations; operations of systems; aircraft systems; and principles of flight--engine inoperative. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Oral Certificates and Documents Airworthiness Requirements Performance and Limitations Operations of Systems Aircraft Systems Principles of Flight--Engine Inoperative
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** have adequate knowledge of all tasks listed and will perform each task to the completion standards set forth in the current FAA Commercial Pilot Airman Certification Standards (AMEL).

Lesson 7-STF–Stage Check

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This Lesson provides a review of normal, abnormal, and emergency operations, procedures, and maneuvers. The Lesson will be conducted by the Chief/Assistant Chief or Designated Check Pilot.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management Automation Management
Preflight Procedures Lesson Planning Preflight Briefing
Cockpit Familiarization Flight Controls Engine Controls
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Flows Cockpit Flows
Engine Start General Starting Procedures Engine Fire On Start Procedures
Avionics GPS Setup and Usage
Taxi Operations Basic Aircraft Control Nose Wheel Steering Differential Braking
Before Takeoff Check Engine Check Instrument Check
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff
Climbs Constant Rate Climbs Constant Airspeed Climbs

Operation of Systems In-Flight Powerplant Operation Use of Constant Speed Propeller Use of Retractable Landing Gear Use of Cowl Flaps Mixture Leaning including use of EGT
Emergency/Abnormal Procedures Engine Failure During Takeoff Before Vmc (Simulated) Engine Failure After Lift-Off (Simulated) Systems and Equipment Malfunctions Mixture Leaning including use of EGT Emergency Descent
Steep Turns Steep Turns
Multiengine Operations Maneuvering With One Engine Inoperative Vmc Demonstration
Slow Flight Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents
Stalls Power-Off Stalls Power-On Stalls
VFR Navigation E6B Usage Diversion Procedures
Cross Country Flight Landing at an Airport at least 50 nm distance
Descents Constant Rate Descents Constant Airspeed Descents

Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Landing With an Inoperative Engine (Simulated) Go-Around/Rejected Landing
Postflight Operations Securing Aircraft Postflight Debrief
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency of basic aircraft avionics, route procedures, aircraft control and operations, and demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 10 degrees and altitude will be maintained within +/- 100 feet. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Stage 2

Stage Objective

The objective of this stage is to further develop the **PUI's** VFR and IFR skills. During this stage, the **PUI** will fly instrument approach procedures, Standard Terminal Arrival Routes (STAR's), Departure Procedures (DP's), holding patterns, and will be able to apply Single Pilot Resource Management (SRM) skills to all areas of flight with little or no input from the **IP**.

The **PUI** will conduct training in the following equipment:

Training Device (TD)
Advanced Aviation Training Device **(AATD)**
Multi-Engine Flight Training Device **(MFTD)**

Aircraft
Airplane Multi-Engine Land **(AMEL)**

Stage Completion Standards

At the completion of this stage, the **PUI** will demonstrate to the **IP** an increased level of proficiency in VFR and IFR cross-country procedures, DP's and STAR's. Additionally, the **PUI** will be exhibiting SRM skills in all phases of flight with little to no input from the **IP**. Importantly, the **PUI** will be introduced to emergency procedures in both the VFR and IFR environment. The **PUI** will be able to accomplish the objectives of Stage 2 to the appropriate level of performance, as outlined in the lesson completion standards.

NOTE

*In the event an **AATD/ FTD** is not available, the training event prescribed can be conducted in the aircraft with the approval from the Chief Instructor/ Asst. Chief or designee.*

Lesson 8-TD

Equipment

AATD/FTD

Lesson Objective

This lesson provides a review of instrument approach charts, as well as instrument navigation, tracking, and procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Principles of Flight--Engine Inoperative	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Short-Field Takeoff	Review
Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review
Turns (IR)	
Standard Rate Turns	Review
Steep Turns	
Steep Turns	Review

Unusual Attitudes (IR) Unusual Attitude /Recovery Procedures	Introduction
Holding Procedures Holding Procedures	Introduction
Instrument Approach Procedures LOC Approach & Procedures	Introduction
Approach and Landing Procedures Normal and/or Crosswind Approach Procedures Go-Around/Rejected Landing	 Review Review
Postflight Operations Securing Aircraft Postflight Debrief	 Review Review
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency in local instrument navigation procedures and will have reviewed instrument approach charts. The **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/-10 degrees, altitude maintained within +/- 100 feet, and airspeed will be maintained within +/- 10 knots. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 9-TD

Equipment

AATD/FTD

Lesson Objective

This lesson provides a review of instrument approach charts, as well as instrument navigation, tracking and procedures and an introduction to single engine instrument approach procedures. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Principles of Flight--Engine Inoperative	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review

Emergency/Abnormal Procedures	
Engine Failure During Takeoff Before Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Systems and Equipment Malfunctions	Review
Mixture Leaning including use of EGT	Review
Emergency Equipment and Survival Gear	Review
Emergency Descent	Review
Multiengine Operations	
Maneuvering With One Engine Inoperative	Review
Instrument Approach Procedures	
LOC Approach & Procedures	Review
Single Engine Instrument Approach Procedures	Introduction
Approach and Landing Procedures	
Normal and/or Crosswind Approach Procedures	Review
Inoperative Engine Approach Procedures	Introduction
Go-Around/Rejected Landing	Review
Postflight Operations	
Securing Aircraft	Review
Postflight Debrief	Review
Additional Areas	
As Assigned By Instructor (If Necessary)	
Special Emphasis Areas (If Necessary)	Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency in local instrument navigation procedures and will have reviewed instrument approach charts. The **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table. Heading will be maintained within +/- 10 degrees, altitude maintained within +/- 100 feet, and airspeed will be maintained within +/- 10 knots. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 10-DXC

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This lesson provides a review of instrument approaches, missed approach procedures and holding procedures.

Lesson Content

SRM	
Risk Assessment	Review
Automation Management	Review
Preflight Procedures	
Principles of Flight--Engine Inoperative	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review
Emergency/Abnormal Procedures	
Engine Failure During Takeoff Before Vmc (Simulated)	Review
Engine Failure After Lift-Off (Simulated)	Review
Systems and Equipment Malfunctions	Review
Emergency Descent	Review

Steep Turns (IR) Steep Turns	 Review
Multiengine Operations Maneuvering With One Engine Inoperative Vmc Demonstration	 Review Review
Stalls (IR) Power-Off Stalls Power-On Stalls	 Review Review
Slow Flight (IR) Pitch/Airspeed Control Power/Altitude Control Climbs/Turns/Descents	 Review Review Review
Unusual Attitudes (IR) Unusual Attitude /Recovery Procedures	 Review
Descents Constant Rate Descents Constant Airspeed Descents	 Review Review
Holding Procedures Holding Procedures	 Review
Instrument Approach Procedures ILS Approach & Procedures GPS Approach & Procedures Single Engine Instrument Approach Procedures	 Introduction Review Review
Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Landing With an Inoperative Engine (Simulated) Go-Around/Rejected Landing	 Review Review Review Review
Postflight Operations Securing Aircraft Postflight Debrief	 Review Review
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have gained an increased proficiency in local instrument navigation procedures and will have been introduced to instrument approach charts. The **PUI** will demonstrate knowledge of the tasks listed, as outlined in the preceding table.
Heading will be maintained within +/- 10 degrees, altitude maintained within +/- 100 feet, and airspeed will be maintained within +/- 10 knots. Additionally, the **PUI** will exhibit increased SRM skills in all areas. During the postflight debrief, the **PUI** will be assigned specific tasks to prepare for the next flight activity.

Lesson 11-DL

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

This lesson provides a review of instrument approaches, missed approach procedures, and holding procedures.

Lesson Content

SRM	
Risk Assessment	Review
Collision Avoidance	Review
Task Management	Review
Automation Management	Review
Preflight Procedures	
Lesson Planning	Review
Cockpit Management	Review
Principles of Flight--Engine Inoperative	Review
Preflight Briefing	Review
Cockpit Familiarization	
Flight Controls	Review
Engine Controls	Review
Checklist Usage	
Normal Procedures Checklist	Review
Abnormal Procedures Checklist	Review
Emergency Procedures Checklist	Review
Flows	
Cockpit Flows	Review
Engine Start	
General Starting Procedures	Review
Engine Fire On Start Procedures	Review
Taxi Operations	
Basic Aircraft Control	Review
Nose Wheel Steering	Review
Differential Braking	Review
Before Takeoff Check	
Engine Check	Review
Instrument Check	Review
Takeoff and Climb Procedures	
Normal and/or Crosswind Takeoff	Review
Short-Field Takeoff	Review
Climbs	
Constant Rate Climbs	Review
Constant Airspeed Climbs	Review

Operation of Systems In-Flight	
Powerplant Operation	Review
Use of Constant Speed Propeller	Review
Use of Retractable Landing Gear	Review
Use of Cowl Flaps	Review
Mixture Leaning including use of EGT	Review
Emergency/Abnormal Procedures	
GPS Failure in Flight	Review
Identification of Inoperative Engine	Review
Proper Use of Flight Controls	Review
Climbs at Single-Engine Best Rate of Climb Speed (Vyse)	Review
Unintentional Spin Recovery (Knowledge Requirement Only)	Review
Engine Failure Enroute (VR/IR)	Review
Engine Shutdown, Full Feathering, and Restart at Cruise Altitude	Review
Maneuvering with one Engine Inoperative	Review
Systems and Equipment Malfunctions	Review
Emergency Equipment and Survival Gear	Review
Emergency Descent	Review
Turns	
Standard Rate Turns	Review
Slow Flight	
Pitch/Airspeed Control	Review
Power/Altitude Control	Review
Steep Turns	
Steep Turns	Review
Unusual Attitudes (IR)	
Unusual Attitudes /Recovery Procedures	Review
Multiengine Operations	
Vmc Demonstration	Review
Stalls	
Power-Off Stalls	Review
Power-On Stalls	Review
Accelerated Stalls	Review
Descents	
Constant Rate Descents	Review
Constant Airspeed Descents	Review
Holding Procedures	
Holding Procedures	Review
Instrument Approach Procedures	
ILS Approach & Procedures	Review
Single Engine Instrument Approach Procedures	Review
Approach and Landing Procedures	
Normal and/or Crosswind Landing	Review
Short-field Landing	Review
Landing With an Inoperative Engine (Simulated)	Review
Go-Around/Rejected Landing	Review

Postflight Operations Securing Aircraft Postflight Debrief	 Review Review
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)	 Review

Completion Standards

At the completion of this lesson, the **PUI** will have met the approved FAA Airman Certification Completion Standards for the Commercial Pilot Certificate with Airplane Multiengine Class Rating privileges.

Lesson 12-STO–End of Course Stage Check (Oral)

Equipment

Briefing Area

Lesson Objective

The lesson will be conducted by the Chief/Assistant Chief or Designated End of Course Check Pilot. This lesson provides an evaluation of knowledge related to certificates and documents; airworthiness requirements; performance and limitations; operations of systems; multiengine operations; and principles of flight--engine inoperative. Situations requiring the **PUI** to execute emergency/abnormal procedures may be presented by the **IP** at any time during the lesson.

Certificates and Documents Airworthiness Requirements Performance and Limitations Operations of Systems Aircraft Systems Multiengine Operations Principles of Flight--Engine Inoperative
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)

Completion Standards

At the completion of this lesson, the **PUI** will have met the approved FAA Airman Certification Completion Standards for the Commercial Pilot Certificate with Airplane Multiengine Class Rating privileges.

Lesson 13-STF–End of Course Stage Check (Flight)

Equipment

AIRPLANE MULTI-ENGINE LAND (AMEL)

Lesson Objective

The lesson will be conducted by the Chief/Assistant Chief or Designated End of Course Check Pilot. At the end of the flight, the **PUI** will meet the applicable PTS for a Commercial Pilot AMEL.

Lesson Content

SRM Risk Assessment Collision Avoidance Task Management Automation Management
Preflight Procedures Preflight Assessment Cockpit Management Principles of Flight--Engine Inoperative Preflight Briefing
Cockpit Familiarization Flight Controls Engine Controls
Checklist Usage Normal Procedures Checklist Abnormal Procedures Checklist Emergency Procedures Checklist
Flows Cockpit Flows
Engine Start General Starting Procedures Engine Fire On Start
Taxi Operations Basic Aircraft Control Nose Wheel Steering Differential Braking
Before Takeoff Check Engine Check Instrument Check
Takeoff and Climb Procedures Normal and/or Crosswind Takeoff Short-Field Takeoff
Climbs Constant Rate Climbs Constant Airspeed Climbs

Operation of Systems In-Flight Powerplant Operation Use of Constant Speed Propeller Use of Retractable Landing Gear Use of Cowl Flaps Mixture Leaning including use of EGT
Emergency/Abnormal Procedures Engine Failure during takeoff Before VMC (Simulated) Engine Failure after Liftoff (Simulated) GPS Failure in Flight Identification of Inoperative Engine Proper Use of Flight Controls Climbs at Single-Engine Best Rate of Climb Speed (Vyse) Unintentional Spin Recovery (Knowledge Requirement Only) Engine Failure Enroute (VR/IR) Engine Shutdown, Full Feathering, and Restart at Cruise Altitude Maneuvering with one Engine Inoperative Systems and Equipment Malfunctions Emergency Equipment and Survival Gear Emergency Descent
Turns Standard Rate Turns
Slow Flight Pitch/Airspeed Control Power/Altitude Control
Steep Turns Steep Turns
Unusual Attitudes (IR) Unusual Attitudes /Recovery Procedures
Multiengine Operations Vmc Demonstration
Stalls Power-Off Stalls Power-On Stalls Accelerated Stalls
Descents Constant Rate Descents Constant Airspeed Descents
Holding Procedures Holding Procedures
Instrument Approach Procedures ILS Approach & Procedures Single Engine Instrument Approach Procedures
Approach and Landing Procedures Normal and/or Crosswind Landing Short-field Landing Landing With an Inoperative Engine (Simulated) Go-Around/Rejected Landing

Postflight Operations Securing Aircraft Postflight Debrief
Additional Areas As Assigned By Instructor (If Necessary) Special Emphasis Areas (If Necessary)

Completion Standards

At the completion of this Lesson, the PUI will have exceeded the approved FAA Airman Certification Completion Standards for the Commercial Pilot Certificate with Airplane Multiengine Class Rating privileges.

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